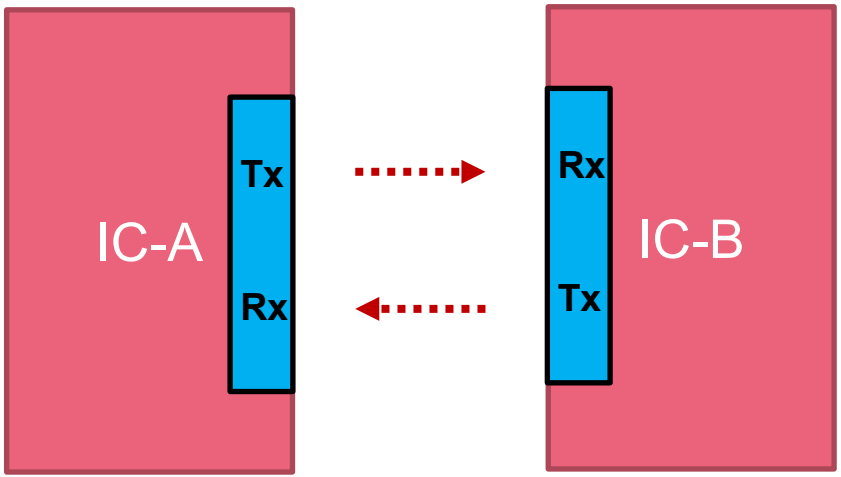
Communication Protocols

UART

UART Protocol

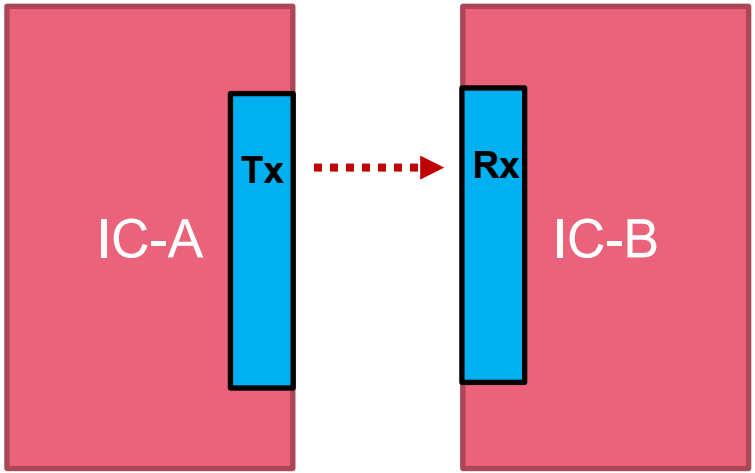
Universal Asynchronous Receiver and Transmitter

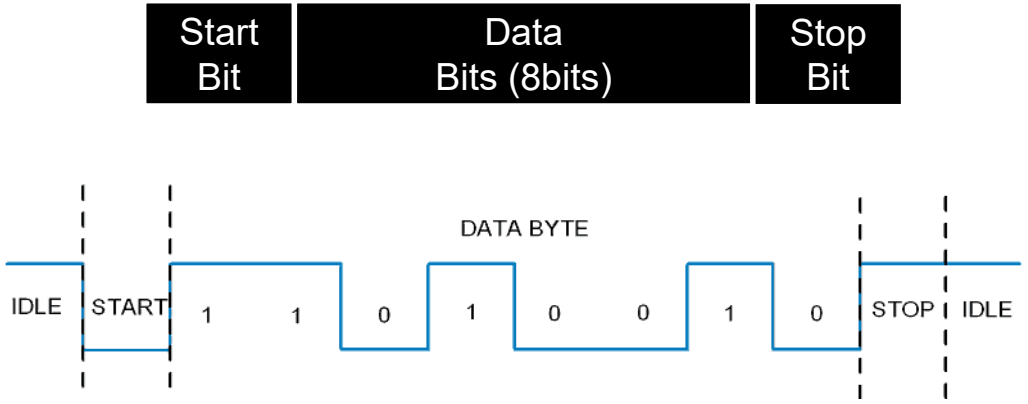


Drive Mode of Pins ?

IC-A to IC-B Pin Connections ?

The Protocol



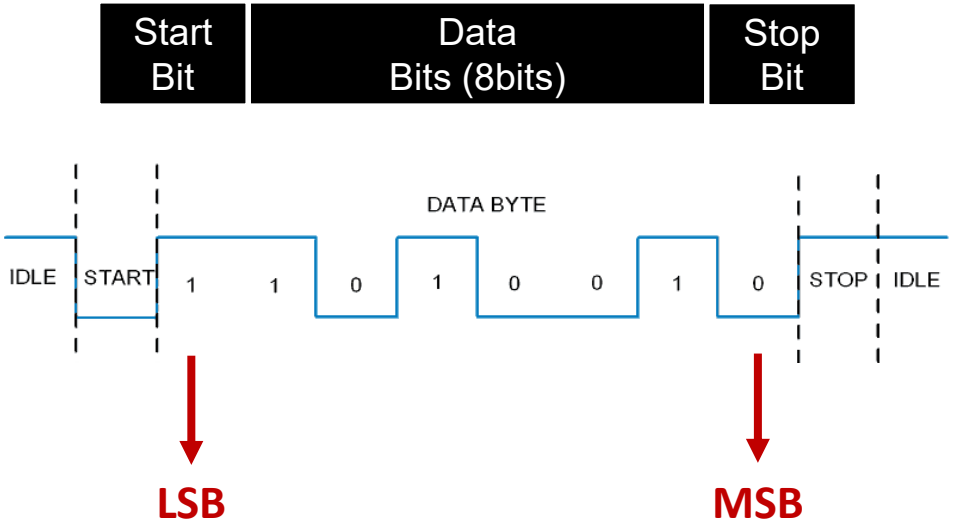


UART baudrate = 1/ 1 bit period

Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

***Mode = 8bit Data Transfer, Parity = None, Stop-Bit=1**

The Protocol

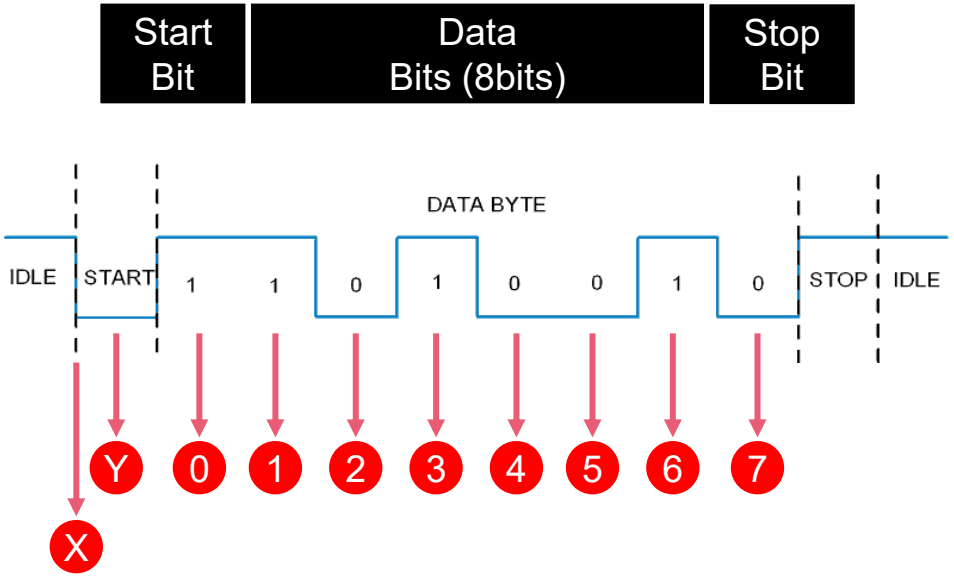


Data byte transferred ?

Why LSB first ?

Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

Decoding

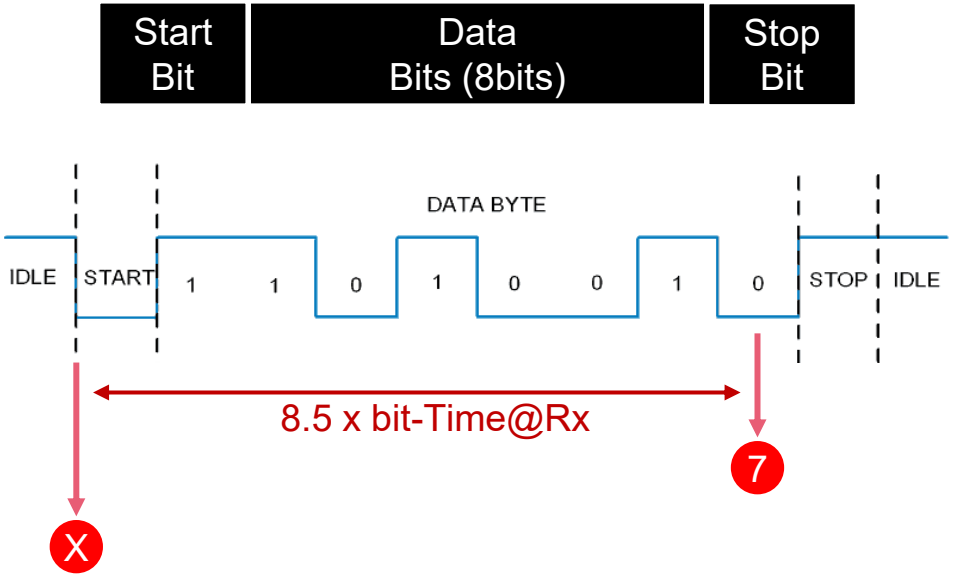


Detects IDLE condition
Y: Wait for 0.5 bit-time
Wait for 1 x bit-time
Sample (bit0)
Wait for 1 x bit-time
Sample (bit1)
…
…
Wait for 1 x bit-time
Sample (bit7)
Wait for 1 x bit-time
Sample (bit8)

Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

What if bit-time@Rx != bit-time@Tx ?

Decoding



If bit-time@Rx != bit-time@Tx ?

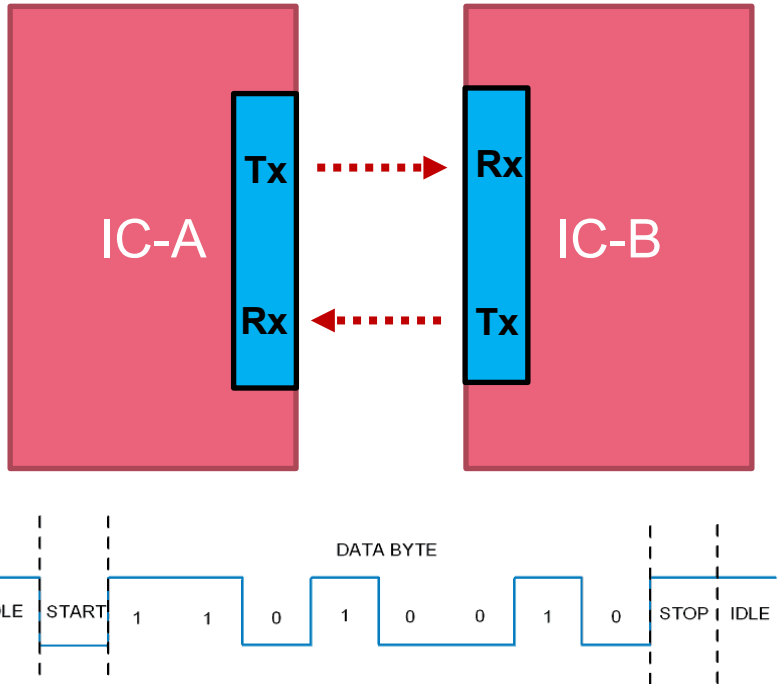
bit-7 is sampled at t = 8.5 x bit-Time@Rx

if 8.5 x bit-Time@Rx > 9 x bit-Time@Tx
Error in sampling bit-7
Samples stop bit instead

if 8.5 x bit-Time@Rx < 8xbit-Time@Tx
Error in sampling bit-7
Samples bit-6 instead

Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

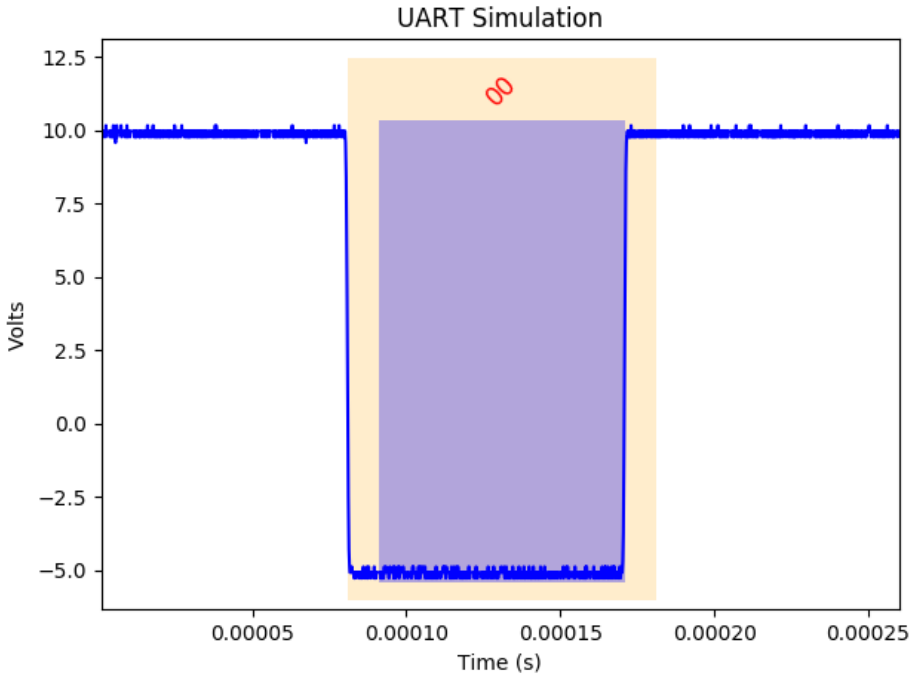
Questions.

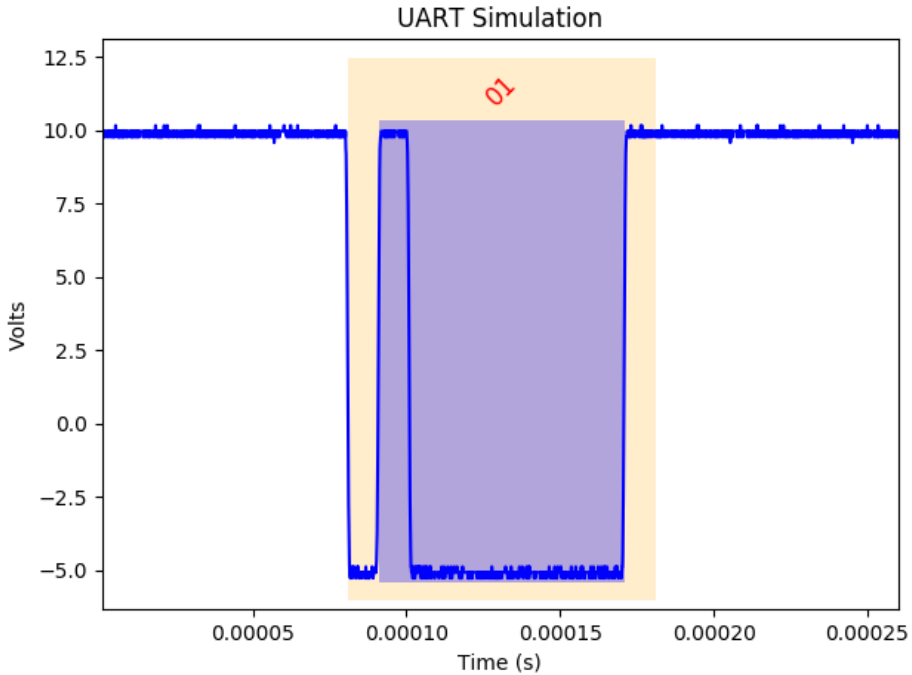


- Significance of baudrate ?
- Need for Start/Stop bit
- How Rx decodes data ?
- Why asynchronous ?
- UART pros/cons ?

Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

Examples



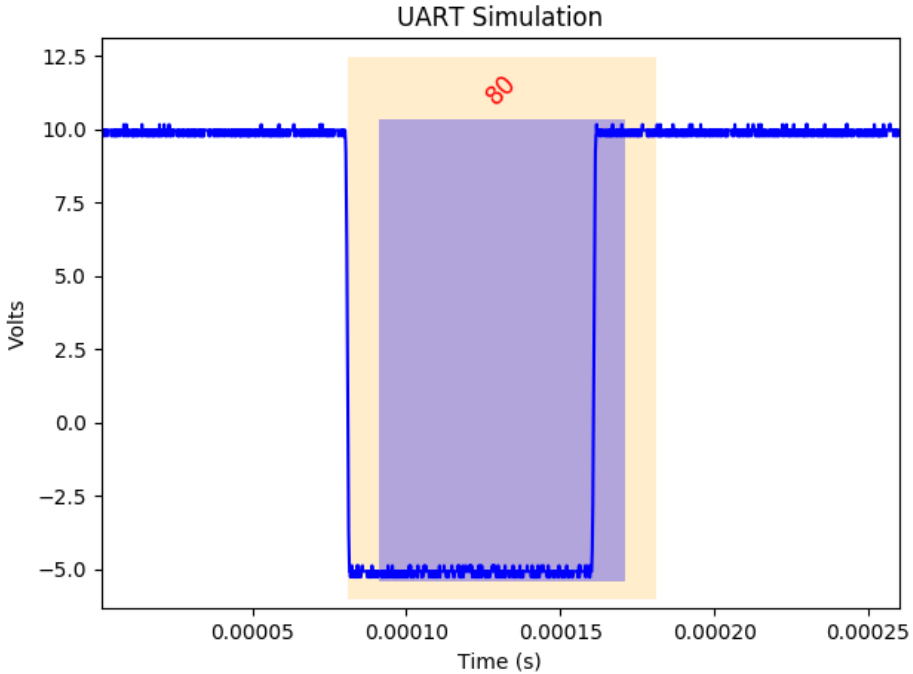


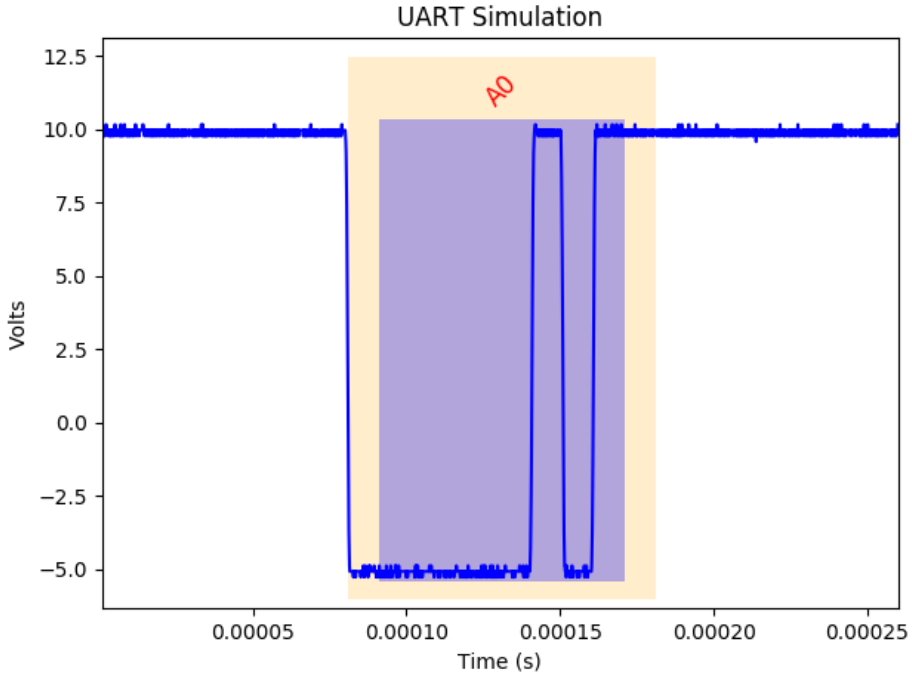
Baudrate ?

Ref – RIPYL Library (Python)

Prof. Kurian Polachan, IIIT-Bangalore

Examples

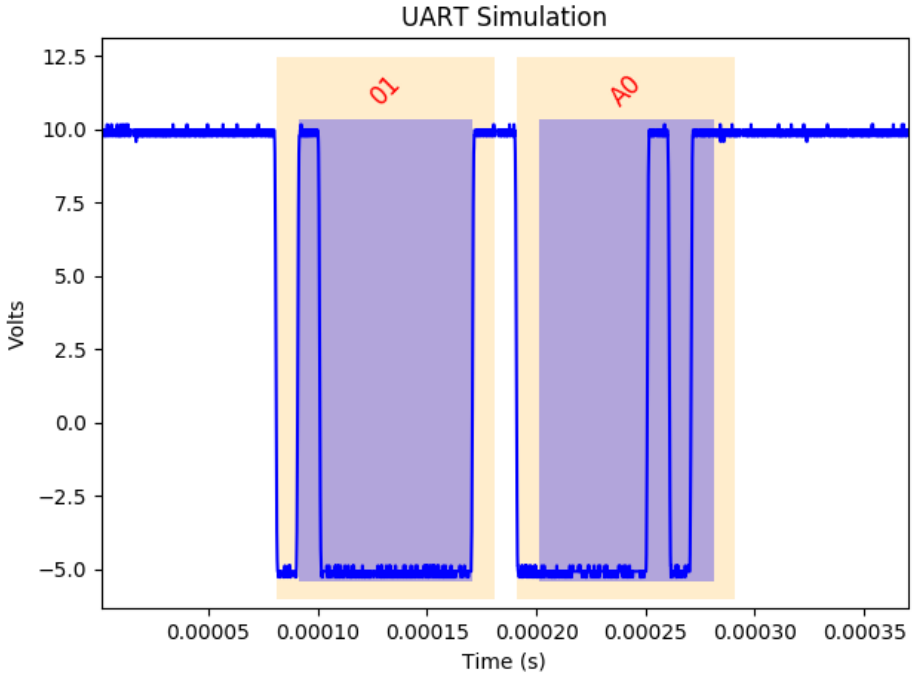




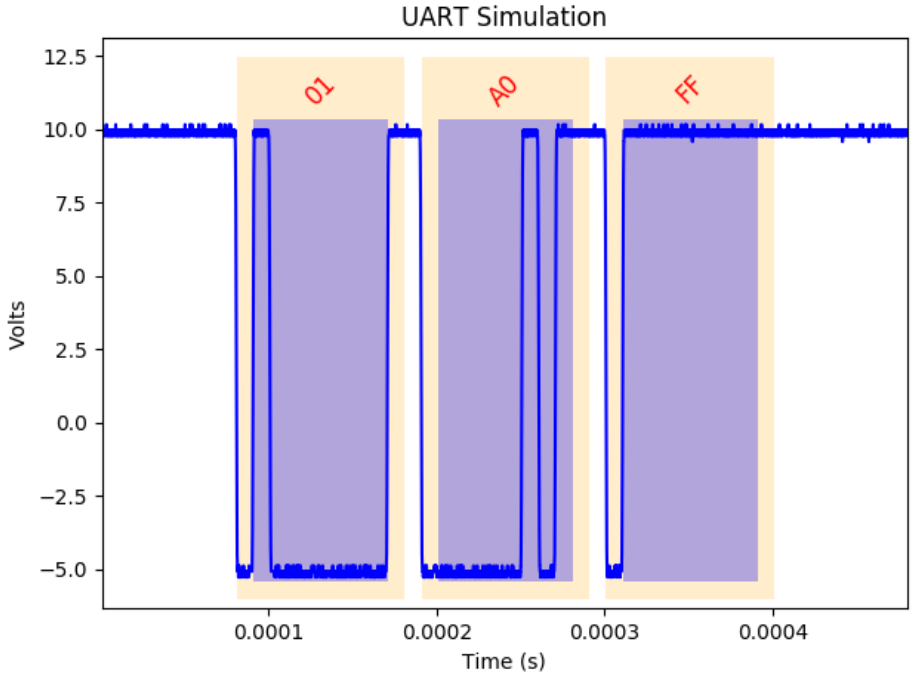
Baudrate ?

Ref – RIPYL Library (Python)

Examples

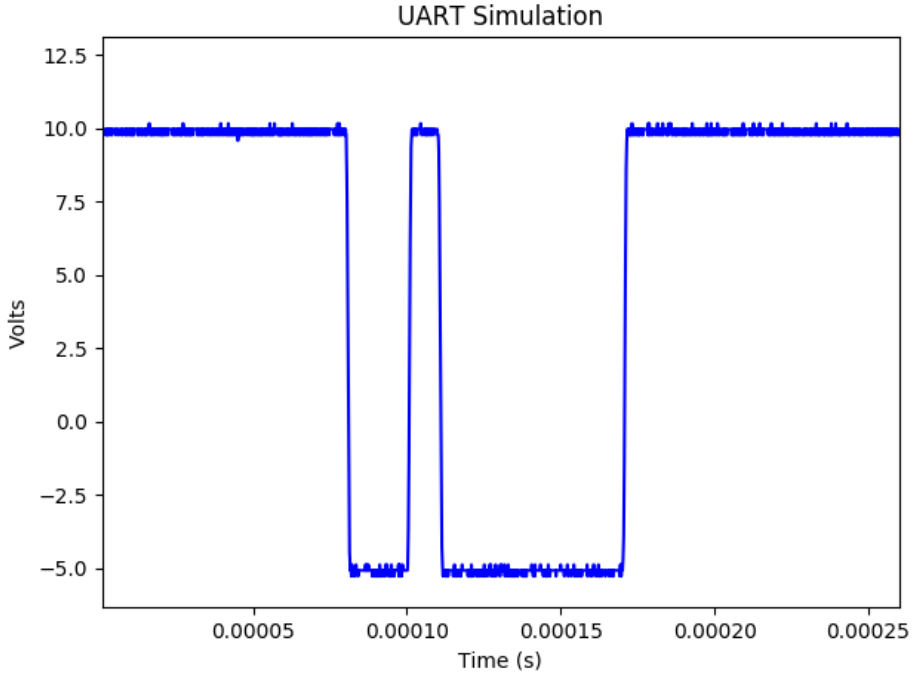


Baudrate ?



Ref – RIPYL Library (Python)

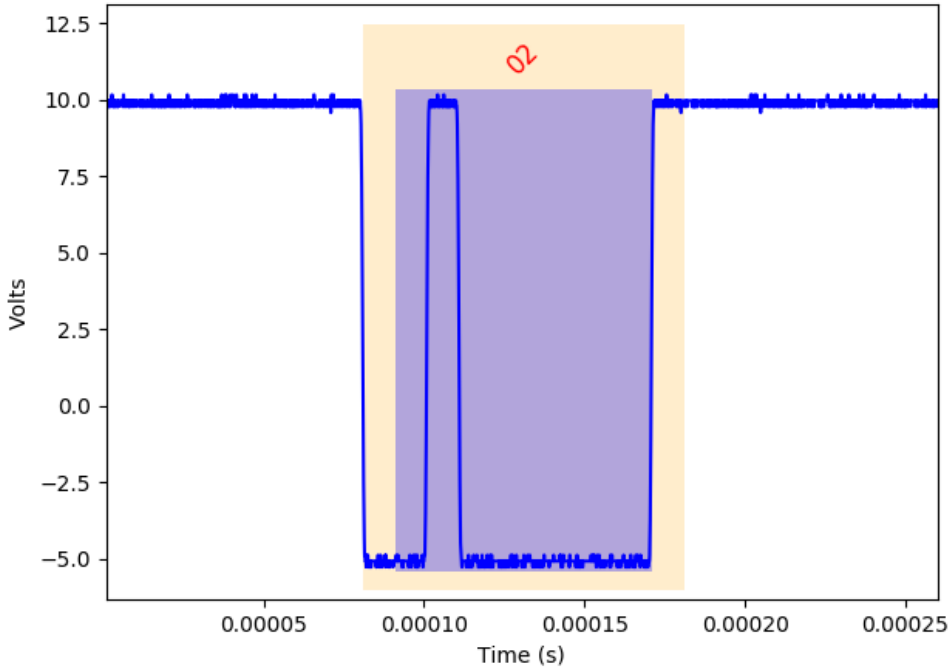
Question.



- **Baudrate ?**
- **Data Byte ?**

Ref – RIPYL Library (Python)

Answer

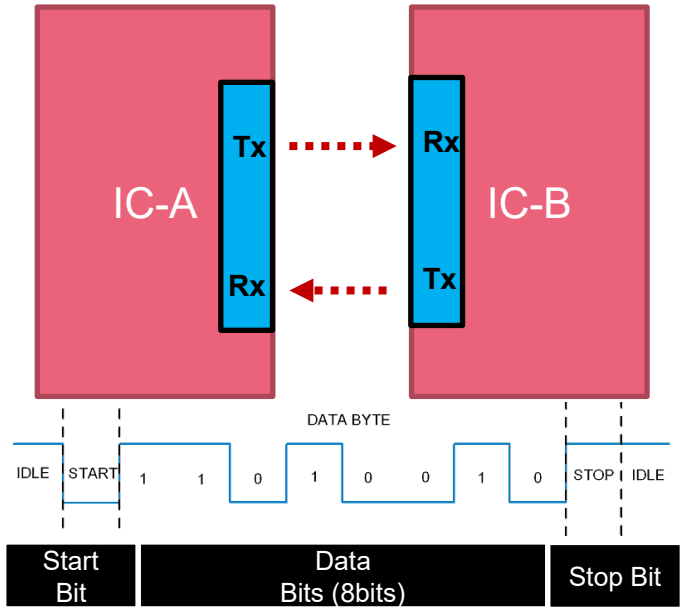


Ref – RIPYL Library (Python)

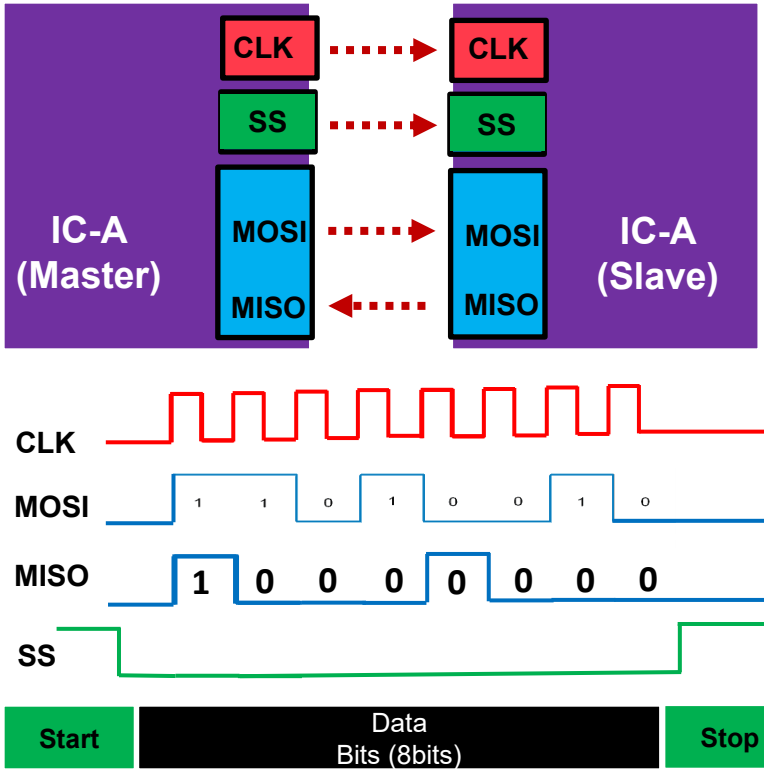
SPI

SPI Protocol

UART

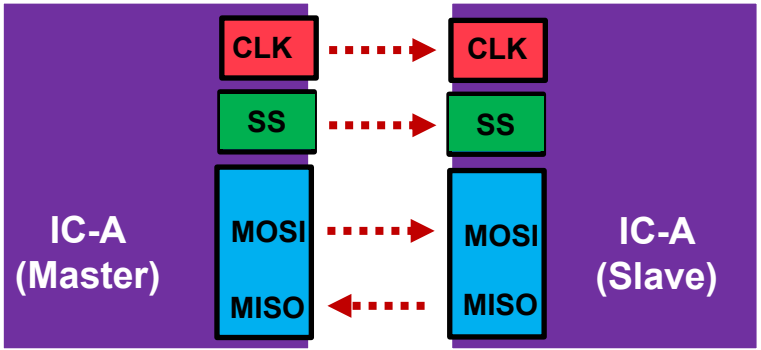


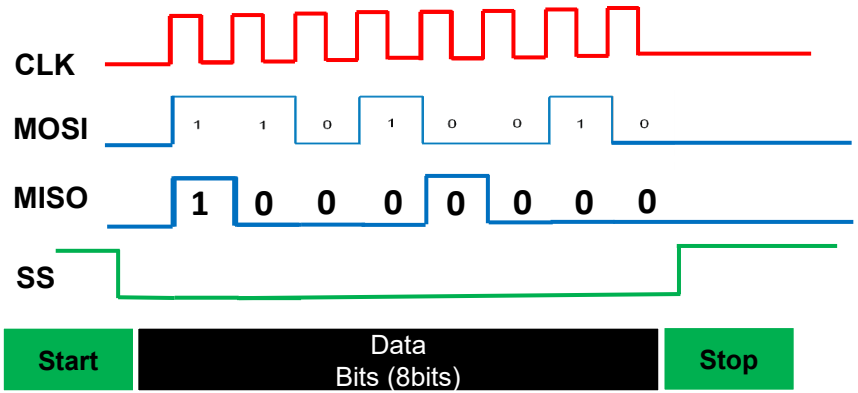
Ref: https://www.electronicwings.com/arm7/lpc2148-uart0

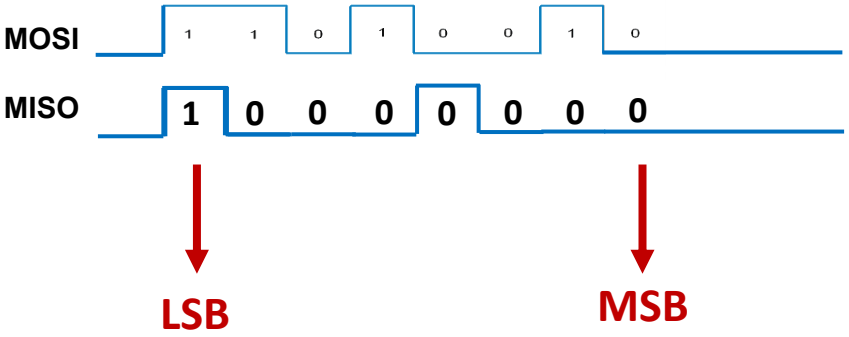


Drive Mode of Pins ?

SPI Protocol



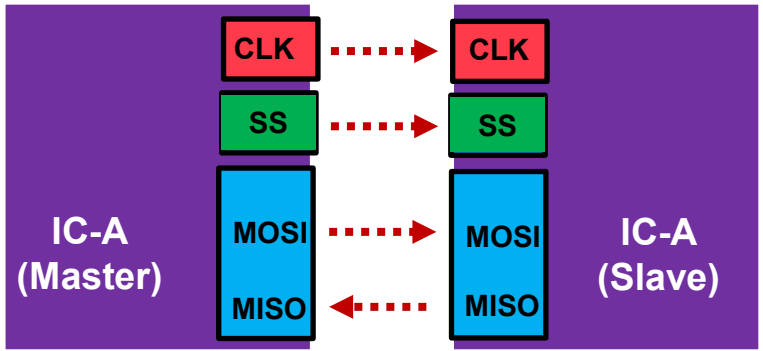


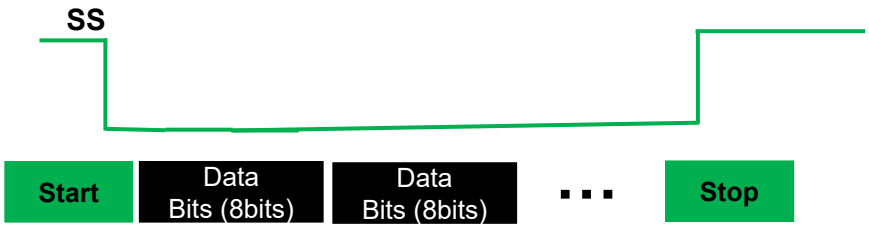


- The clock signal is in the low state when idle.
- The data is changed on the leading edge of the clock
- The data is sampled on the falling edge of the clock

*SPI Mode = 1 (CPOL = 0, CPHA = 1), LSB First Transfer

Multi-Byte Transfer

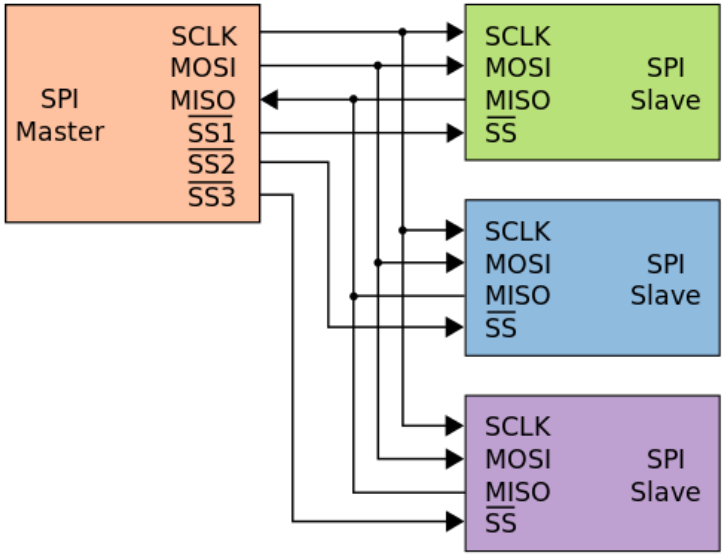




Better than UART

- No Start/Stop Bit Requirement
- Header/Footer Not Required.
- Synchronized. Bits Change with Clock. Transfer starts with SS
- High Speed

Multi-Slave Comm.



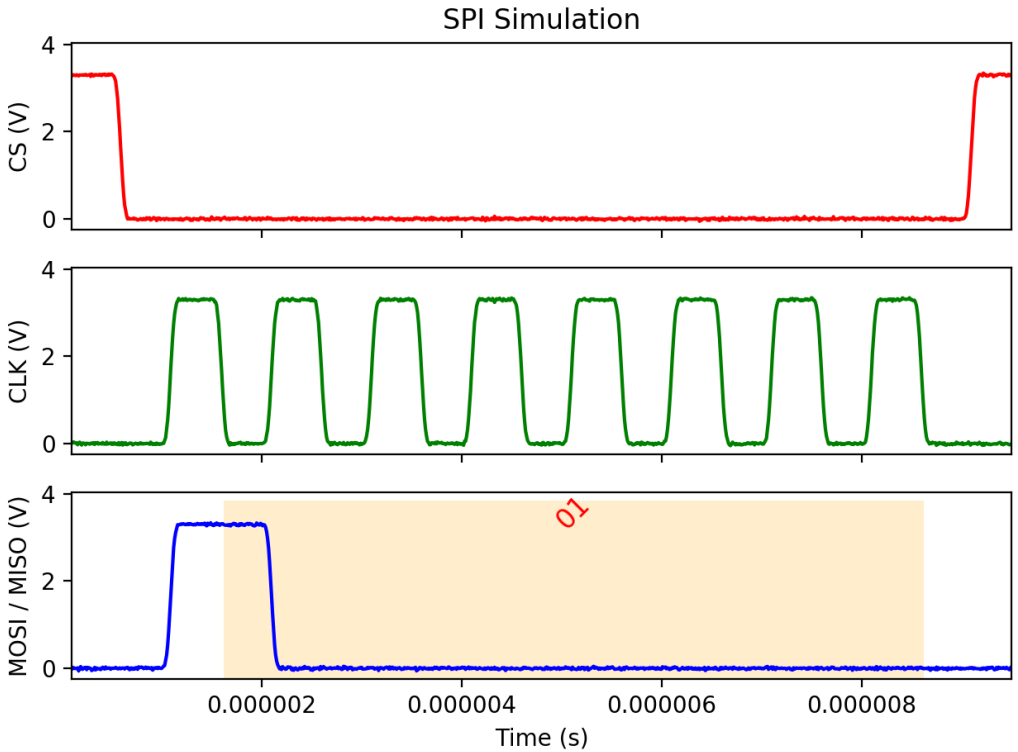
- **How many slave select lines are required for interfacing 8 SPI-Slave devices with SPI-Master**
- **Can UART be used for Multi-Slave Communications ?**

https://en.wikipedia.org/wiki/Chip_select

Questions.

- Significance of SPI Clock Speed ?
- Algorithm for SPI Transmit and SPI Receive ?
- Is SPI Synchronous or Asynchronous ?
- SPI Pros/ Cons. (SPI require more pins ?)

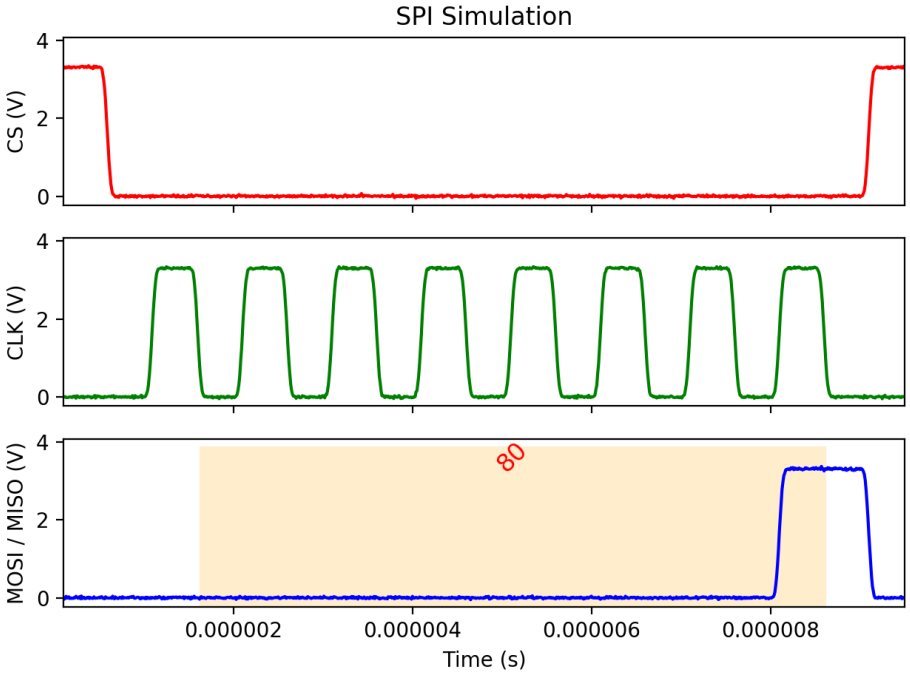
Examples

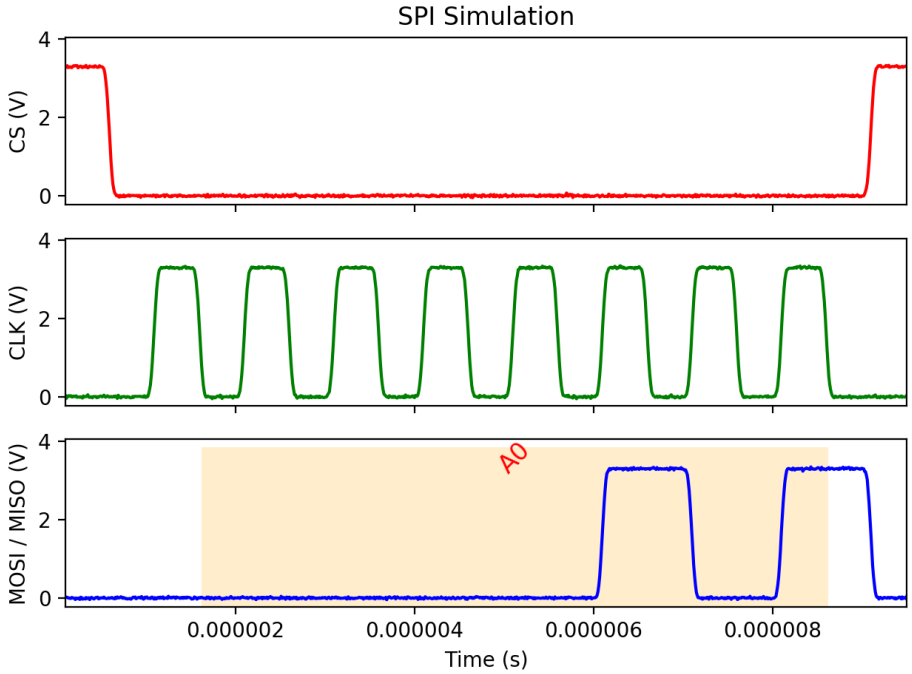


Clock Speed ?

Ref – RIPYL Library (Python)

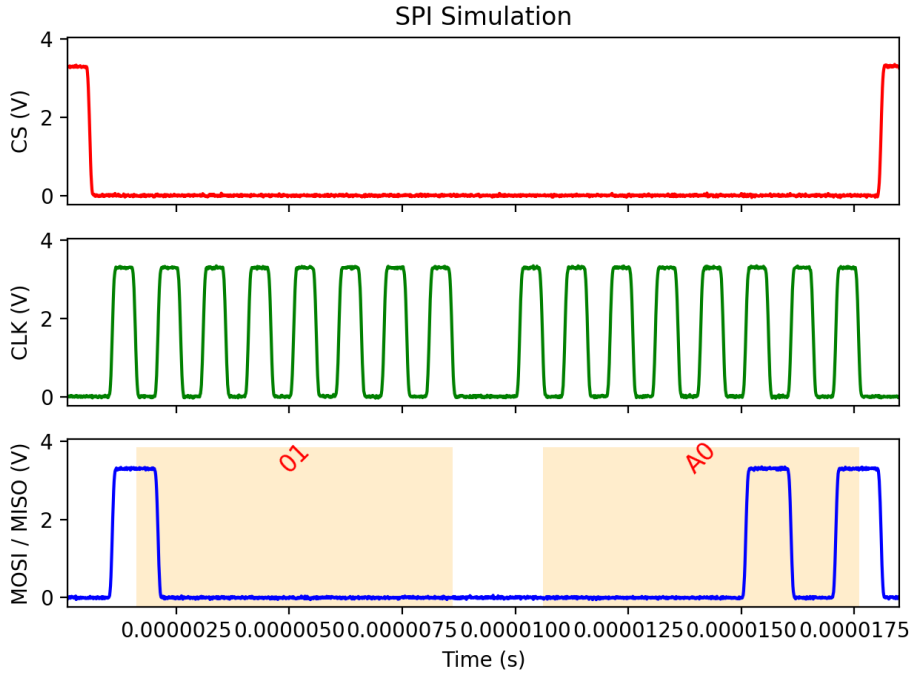
Examples

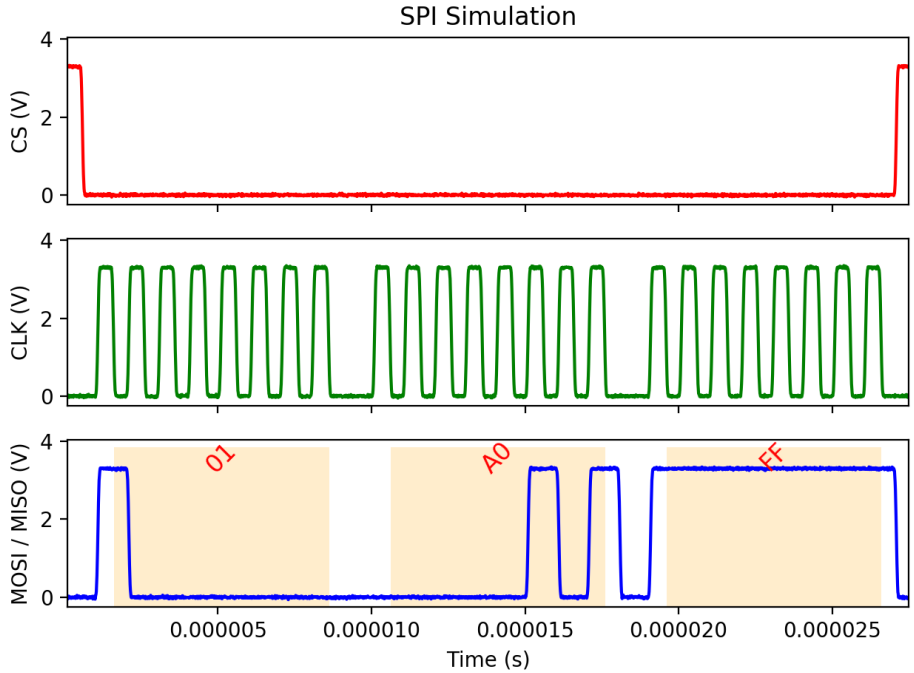




Ref – RIPYL Library (Python)

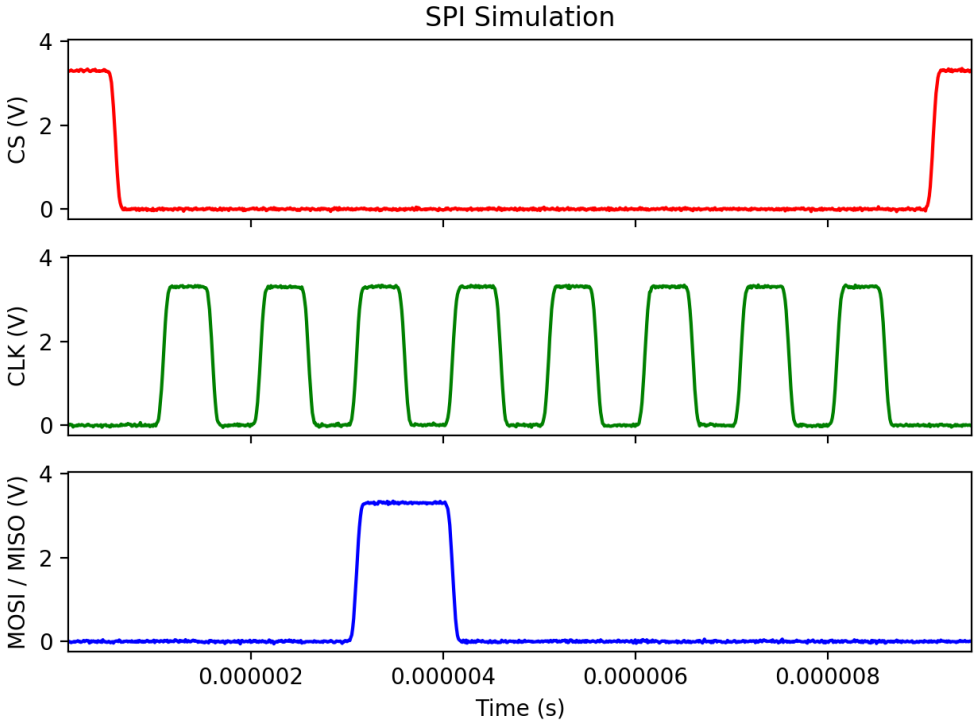
Examples





Ref – RIPYL Library (Python)

Question.

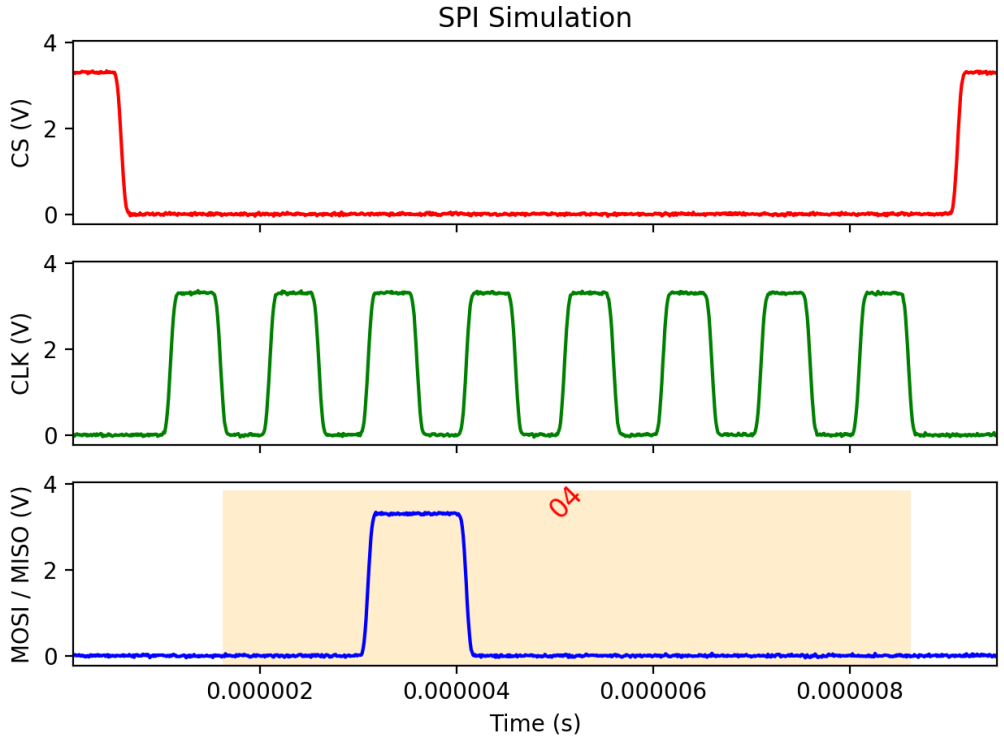


- **Baudrate ?**
- **Data Byte ?**

Ref – RIPYL Library (Python)

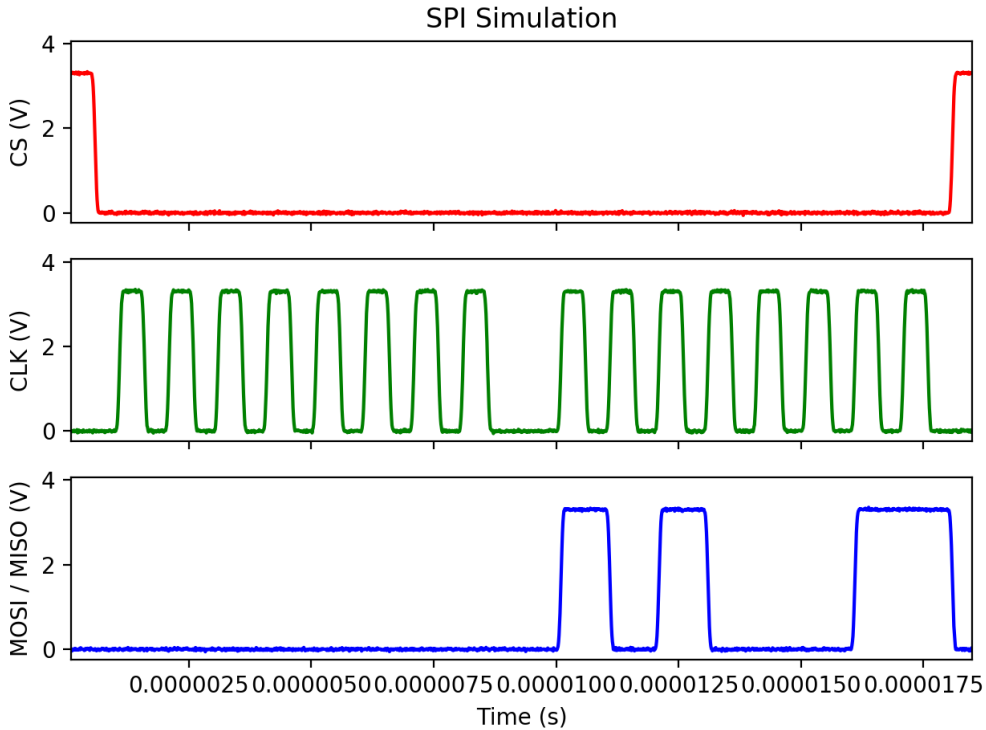
Prof. Kurian Polachan, IIIT-Bangalore

Answer



Ref – RIPYL Library (Python)

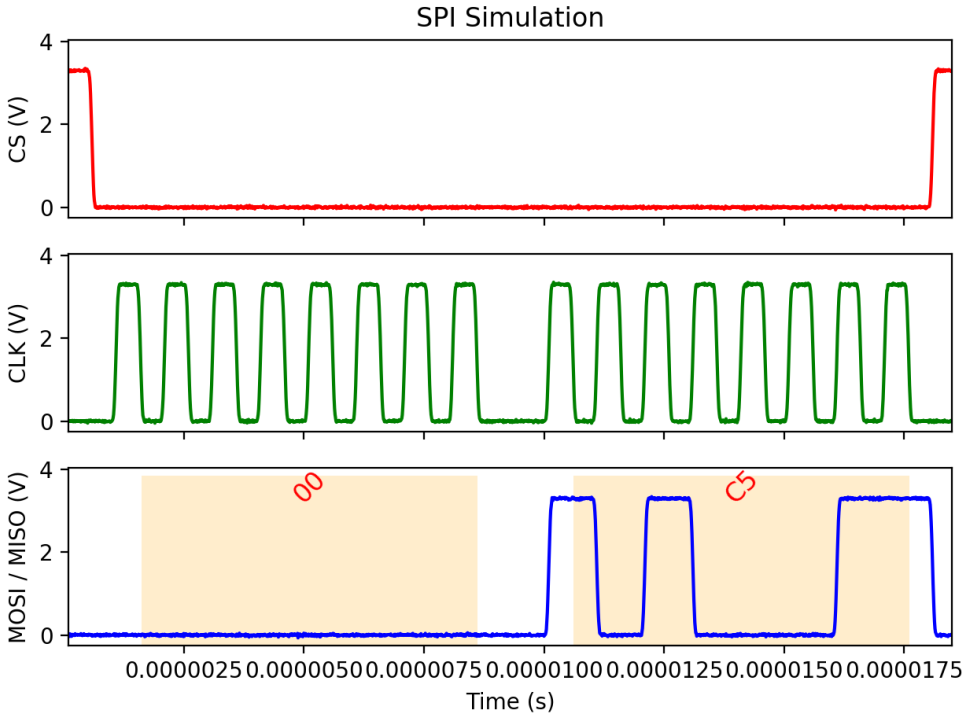
Question.



- **Baudrate ?**
- **Data Byte ?**

Ref – RIPYL Library (Python)

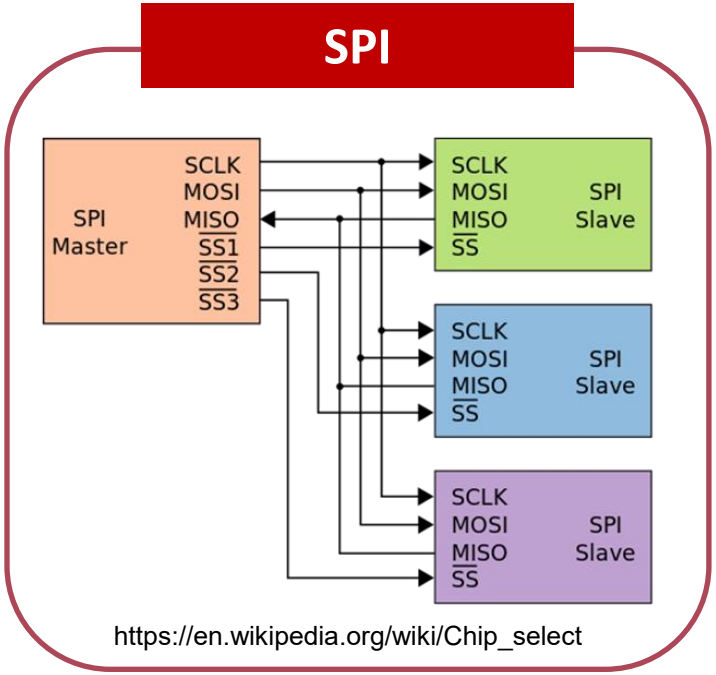
Answer

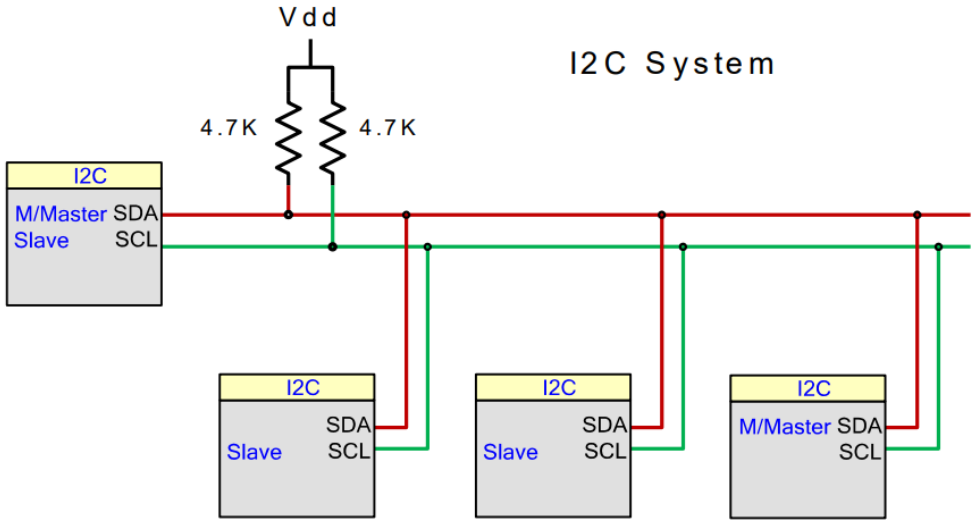


Ref – RIPYL Library (Python)

I2C

I2C Protocol



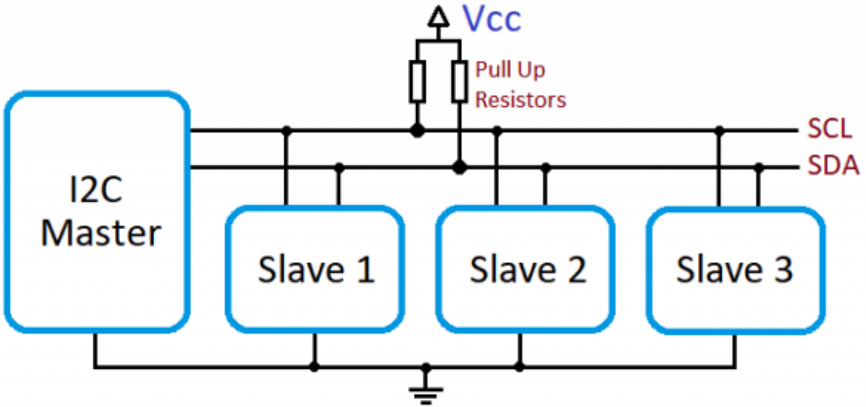


Drive Mode of Pins ?

How to decide Pull-Up Value ?

ref: I2C Master/Slave PSoC Creator Component Data Sheet

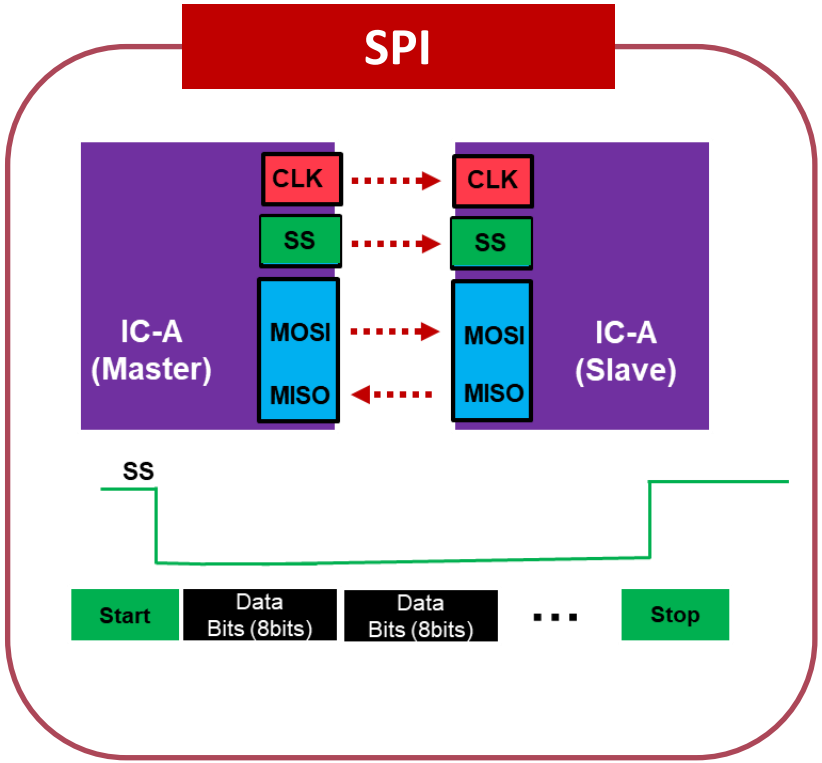
Questions

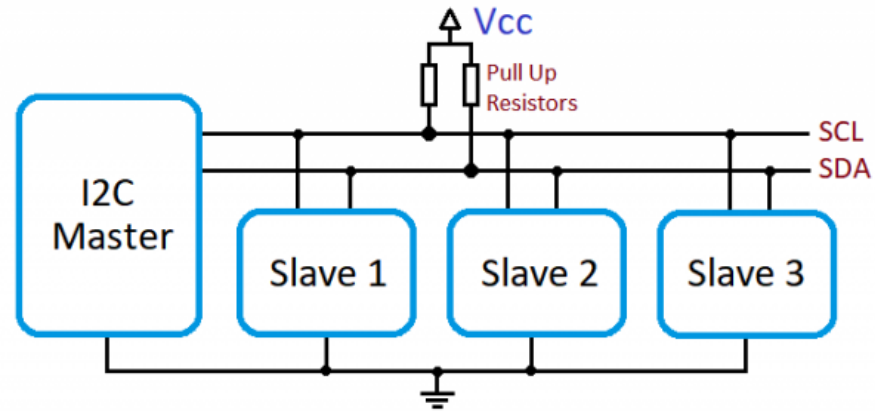


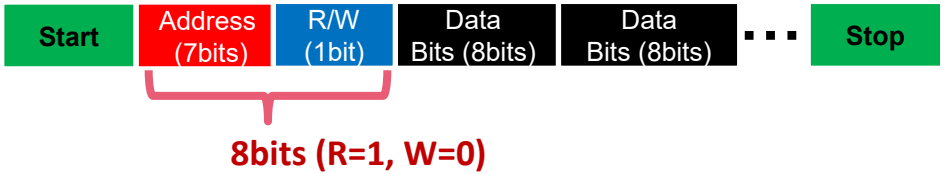
https://electrosome.com/i2c/

- How to Address a Slave ?
- How to transfer data ?
- Why SCL/SDA has pull-up ?

Addressing, Data Tx-Rx

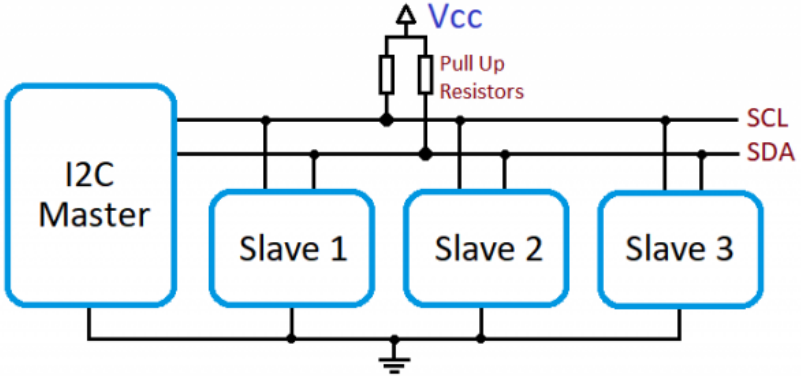




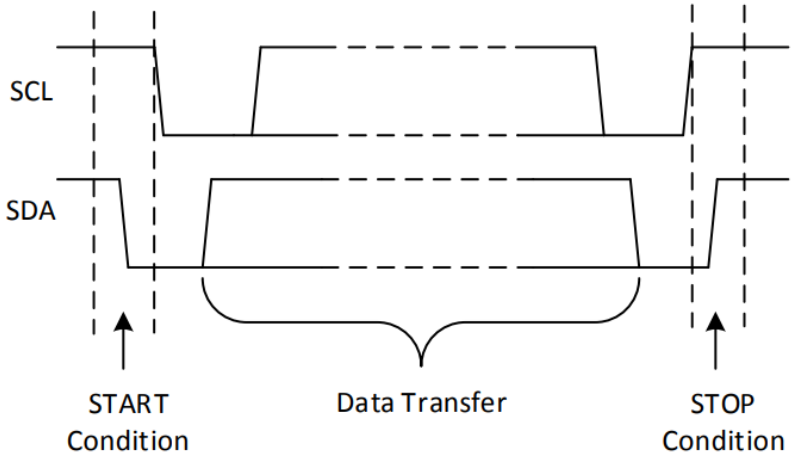


https://electrosome.com/i2c/

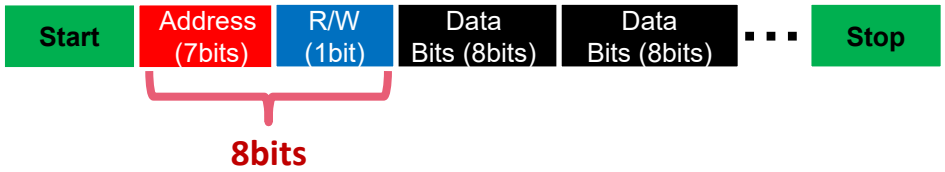
Start/ Stop



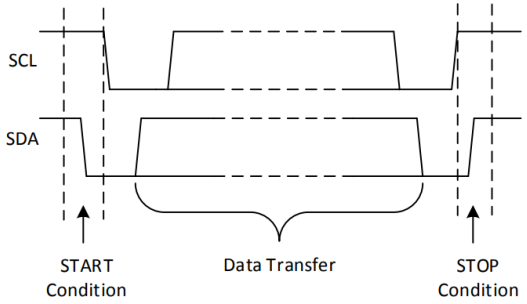
https://electrosome.com/i2c/



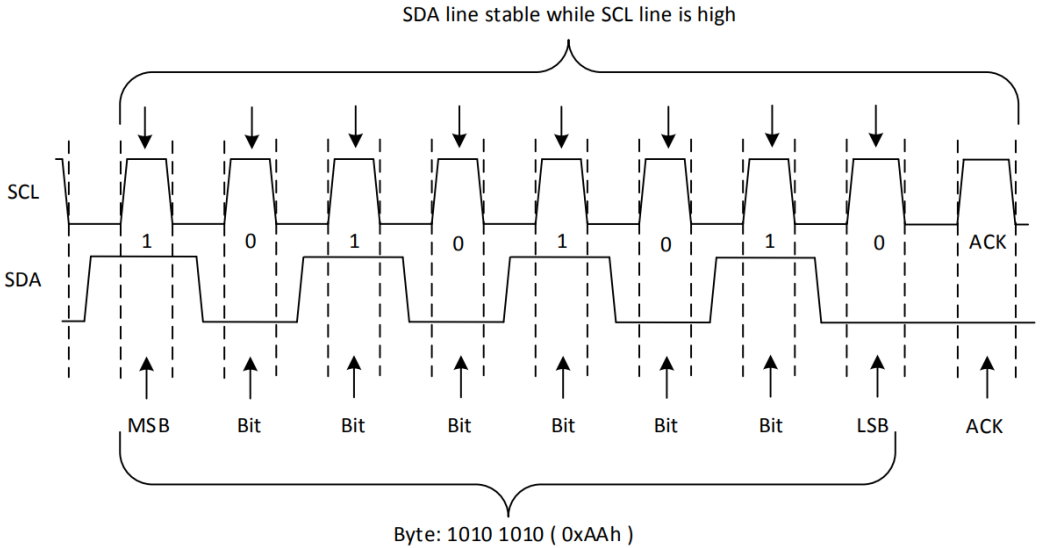
Ref: Understanding the I2C Bus, Texas Instruments

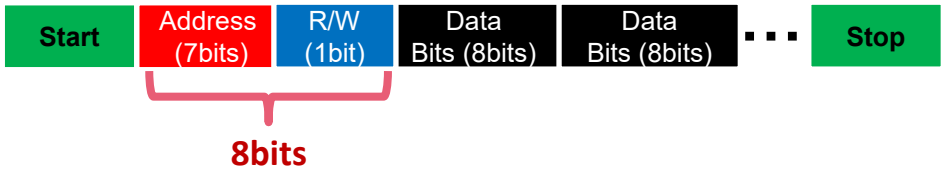


Byte Transfer



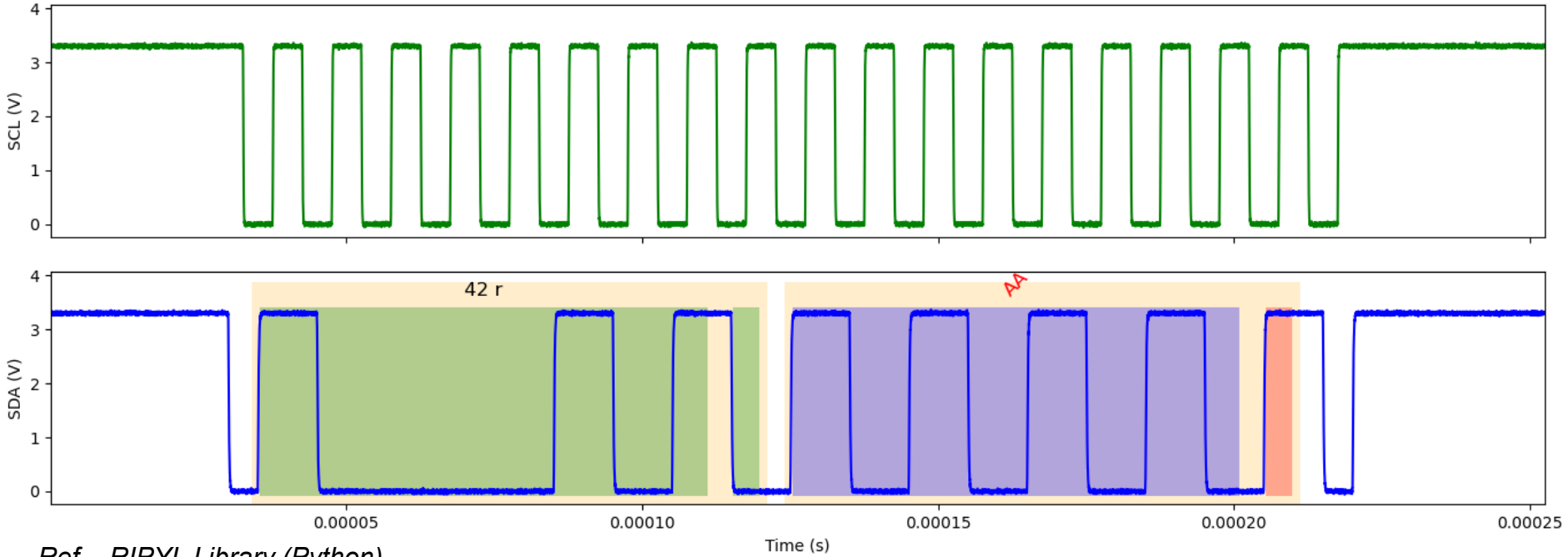
Ref: Understanding the I2C Bus, Texas Instrun





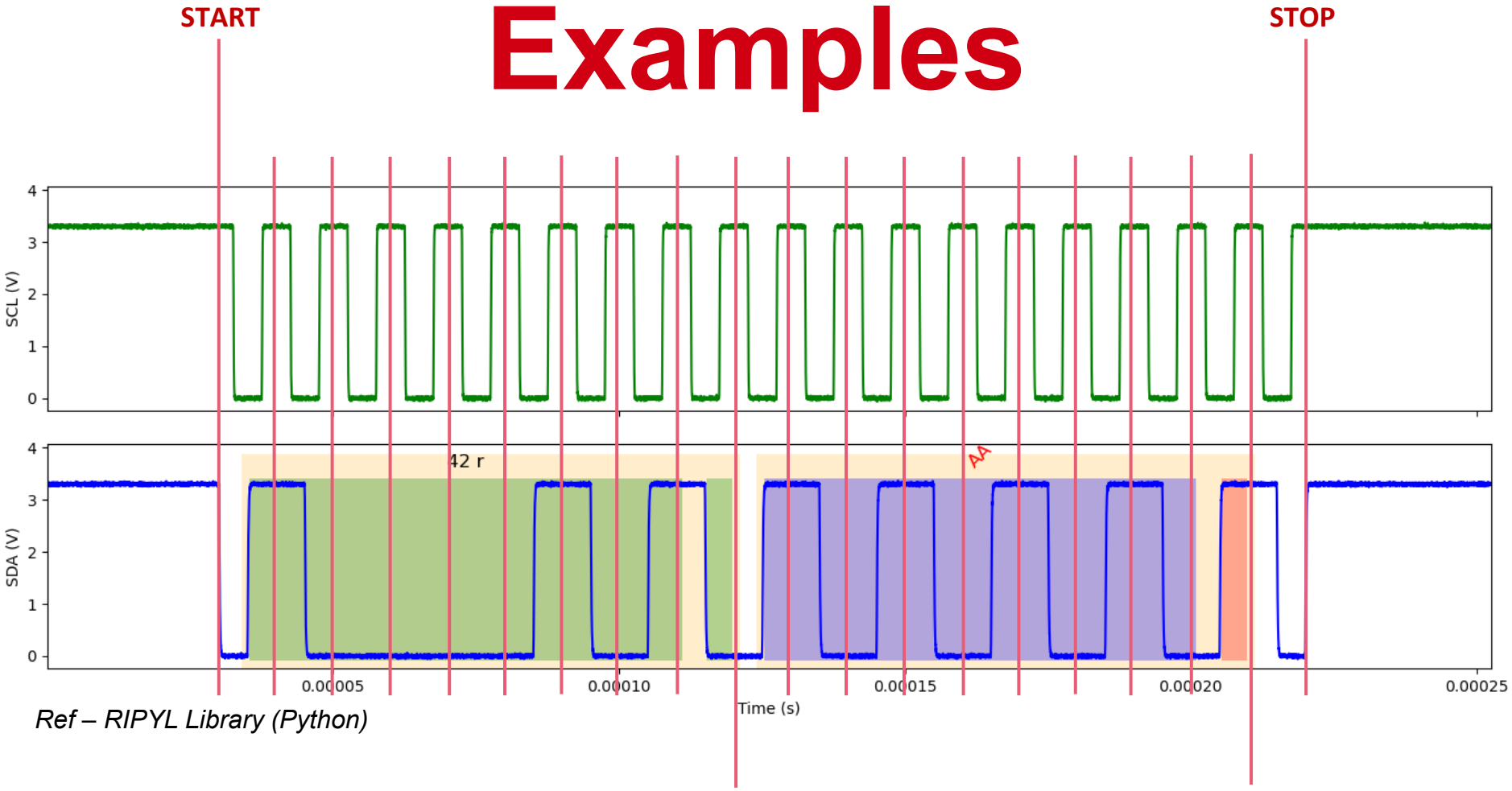
Master expects an acknowledgment from slave after every byte transfer !

Examples



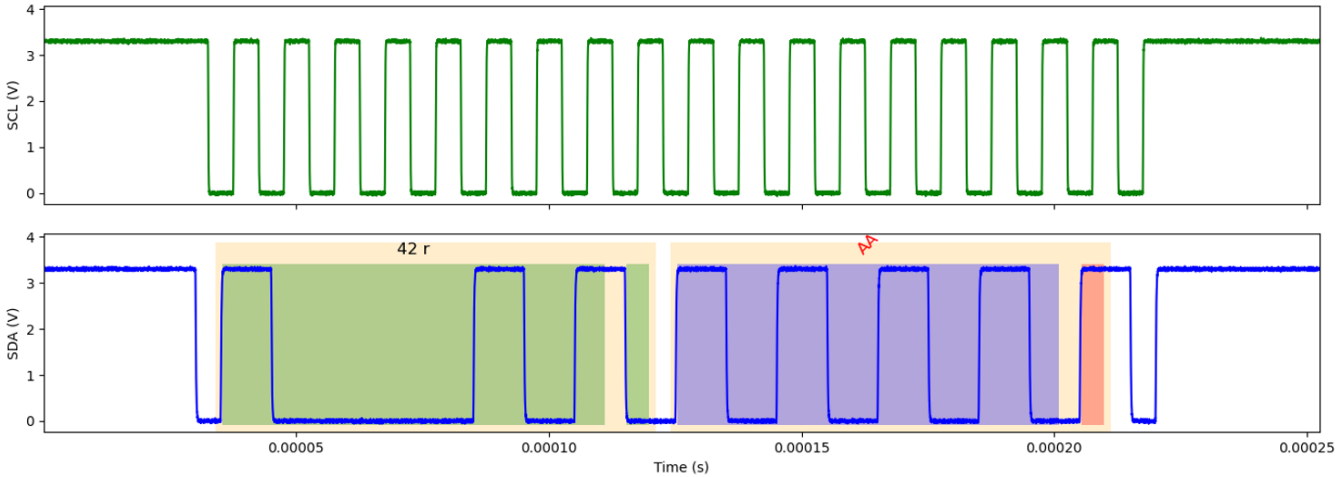
Ref – RIPYL Library (Python)

Examples



Ref – RIPYL Library (Python)

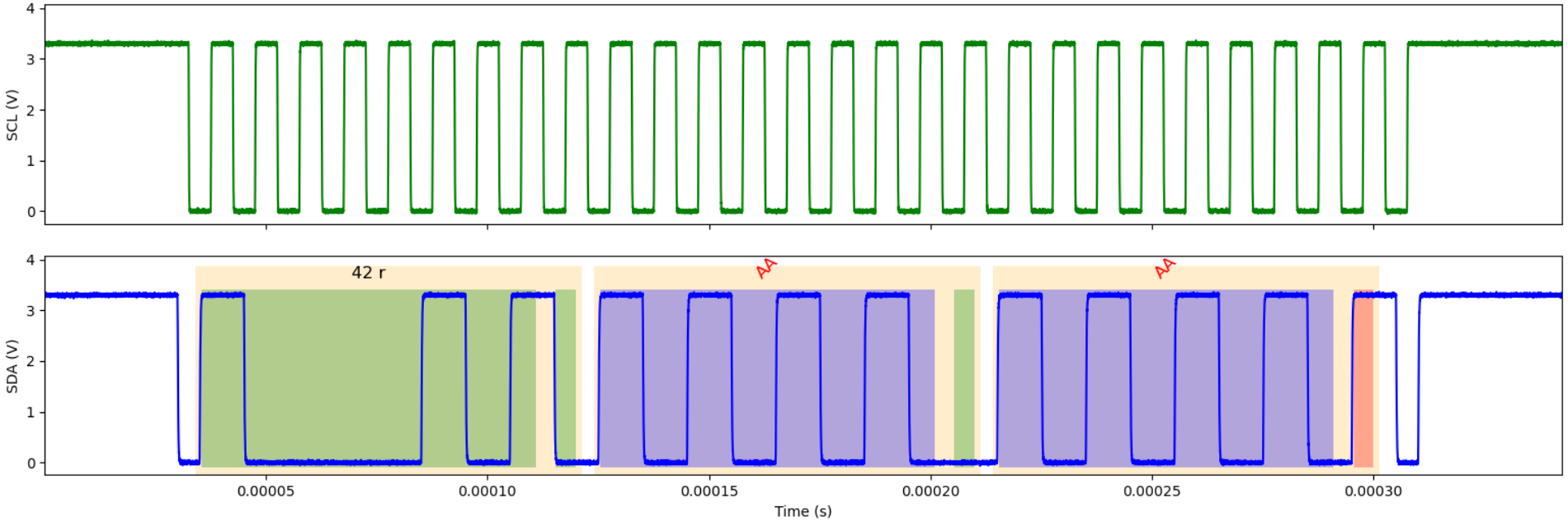
Examples



START
↓
Address + READ (R/W = 1)
↓
Slave ACK
↓
Data byte (e.g., AA)
↓
Master NACK
↓
STOP

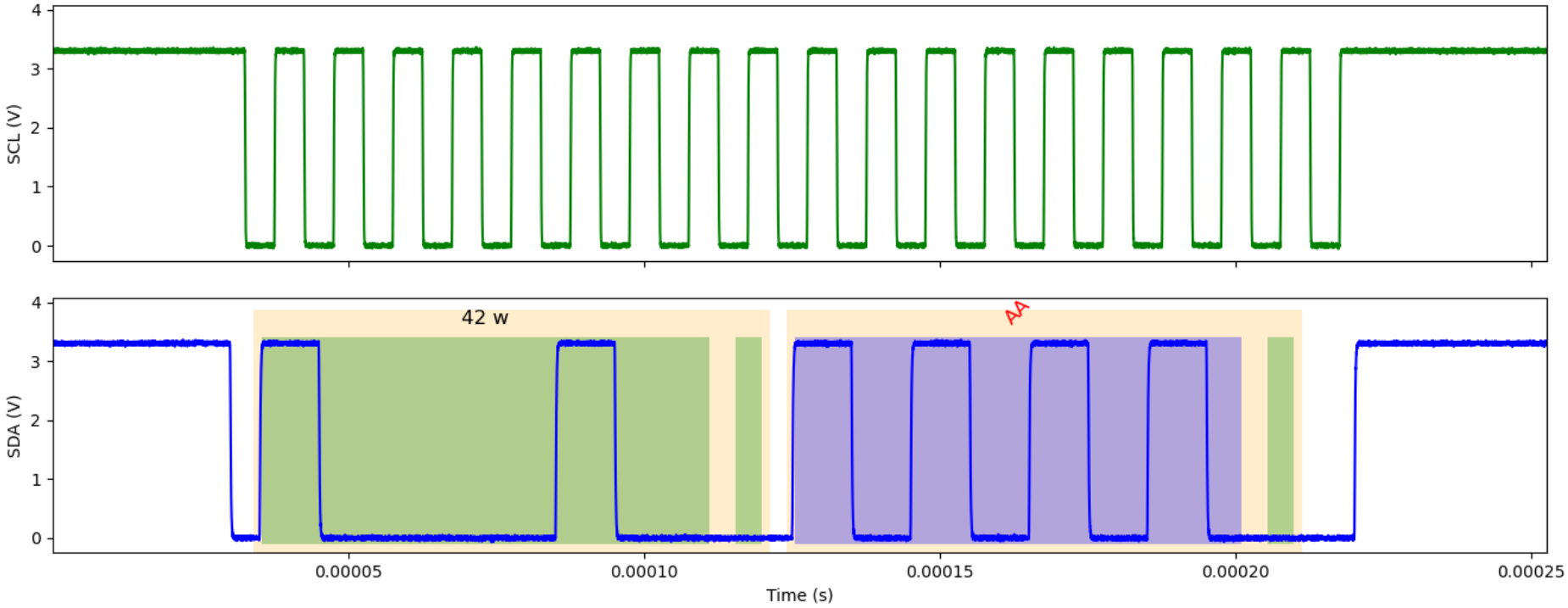
Ref – RIPYL Library (Python)

Examples



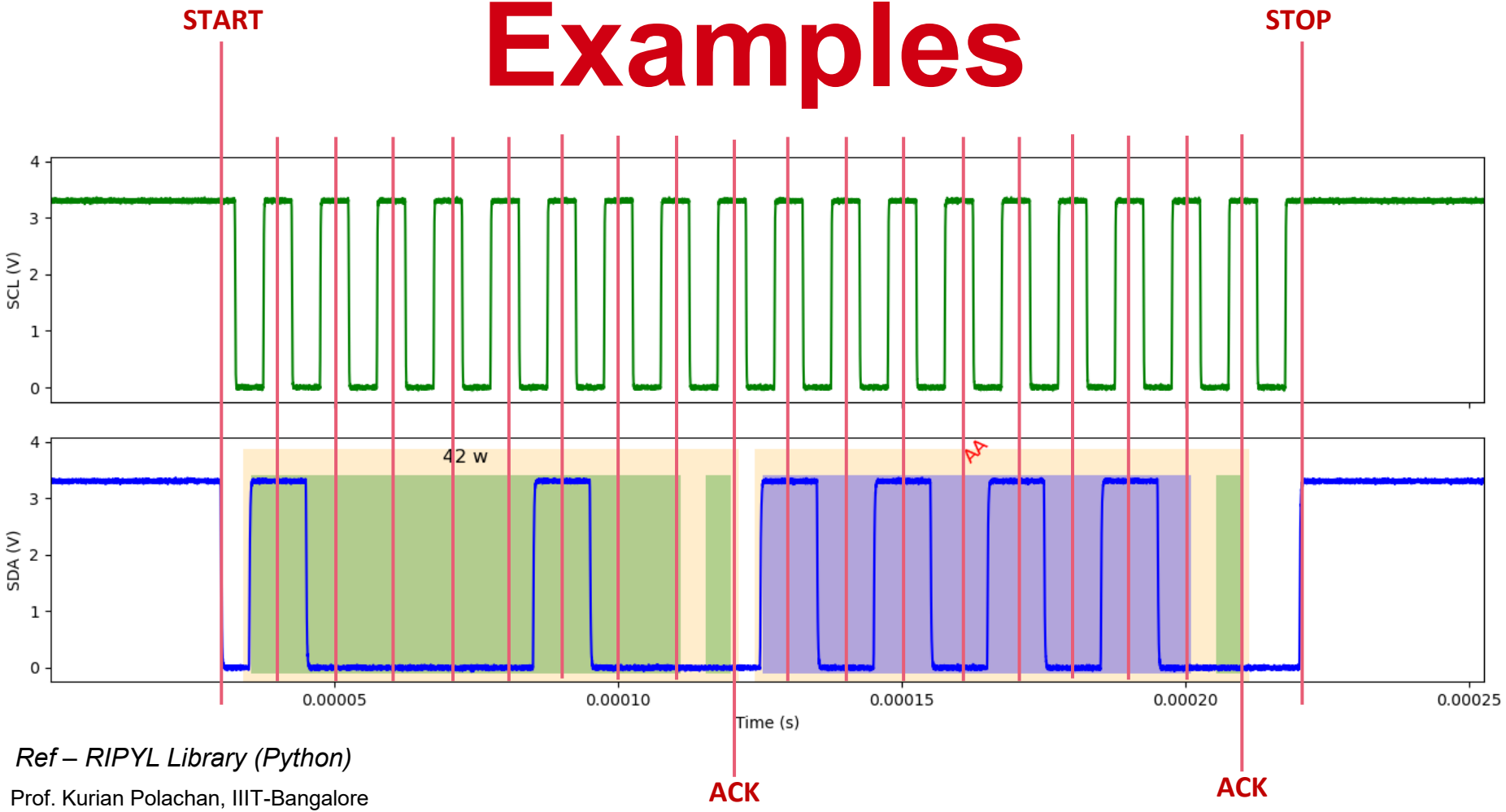
Ref – RIPYL Library (Python)

Examples



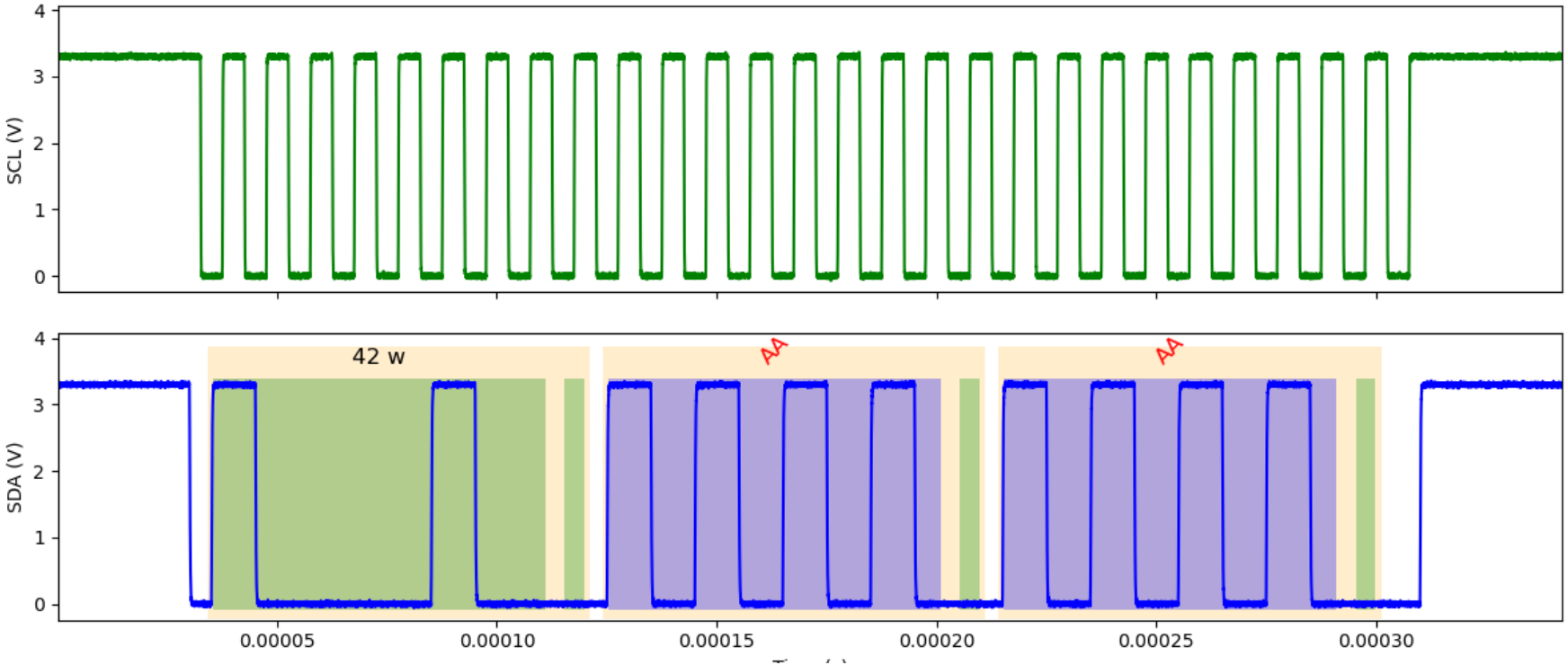
Ref – RIPYL Library (Python)

Examples



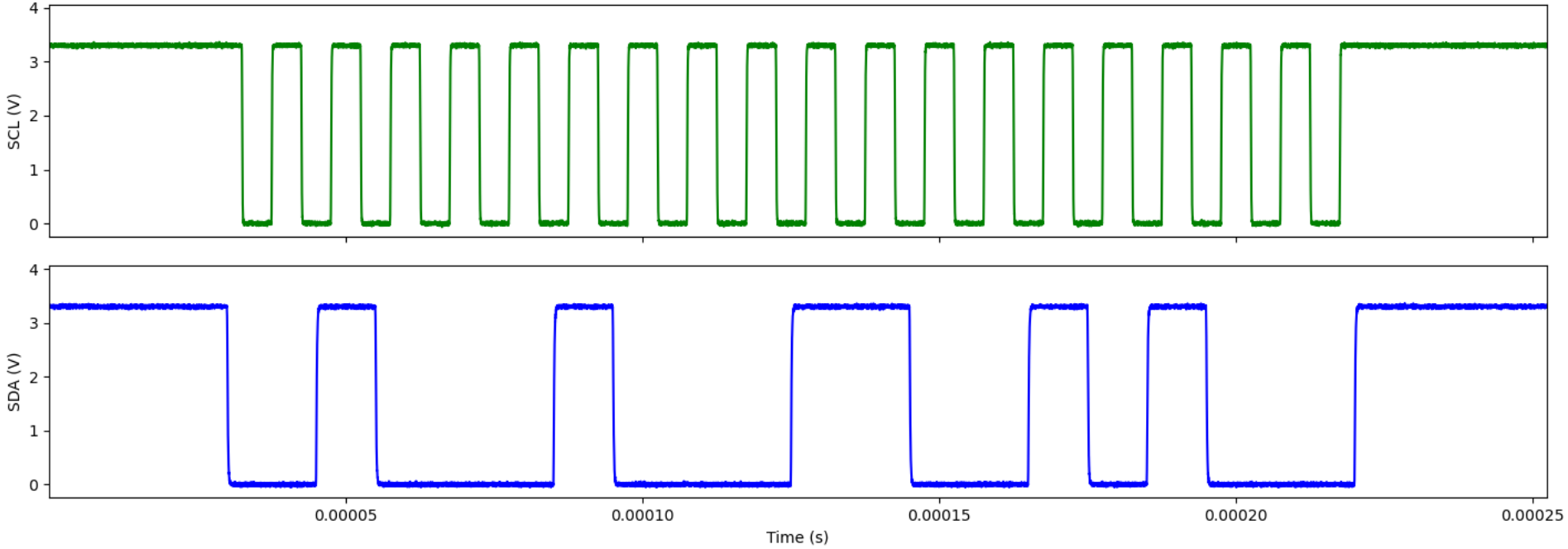
Ref – RIPYL Library (Python)

Examples



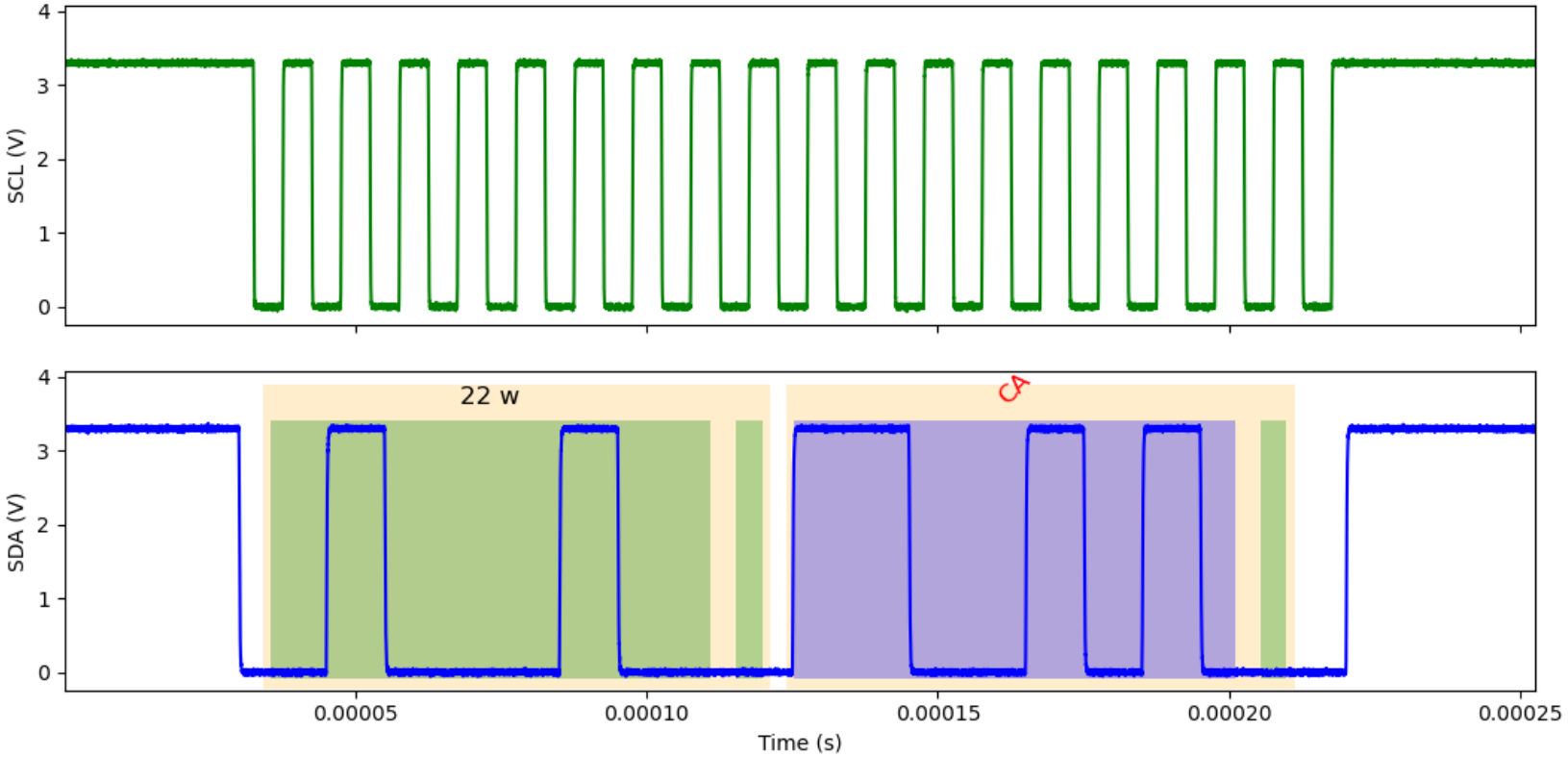
Ref – RIPYL Library (Python)

Question



Ref – RIPYL Library (Python)

Answer



Ref – RIPYL Library (Python)

SPI, I2C and UART

- Simplest Protocol ?
- Slowest Protocol ?
- Multi-Master ?
- Multi-Slave ?
- Detect Presence of Slave ?
- Header/Footer Requirement ?
- Full-Duplex ?
- Synchronous ?